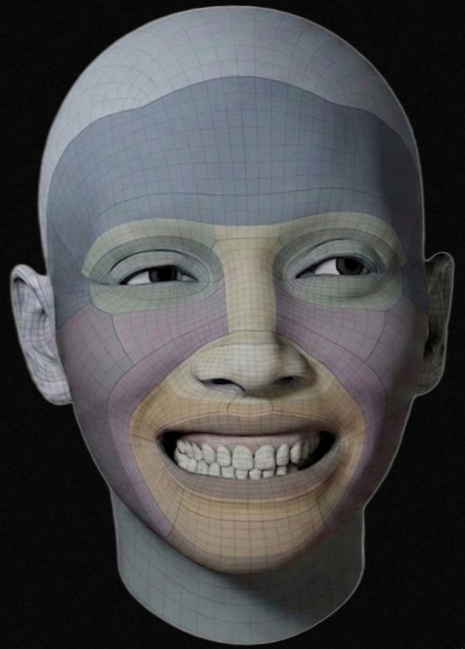
Soutenance finale Etudes Pratiques :

Développer des interactions naturelles grâce au regard avec un agent virtuel en VR

Supervisé par Katja Zibrek



Amadou B. - Tom B. - Maël J. - Thomas L. - Alexandre R.

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{10} QUELLE EST LA SUITE DE CE PROJET ?





01

RAPPELS DU SEMESTRE 5





Agent conversationnel en VR



Regard aléatoire



Conversation complète

Ce qu’on a fait au Semestre 5

02

ÉTAT DE L’ART

Stable



Introverti



Névrosé

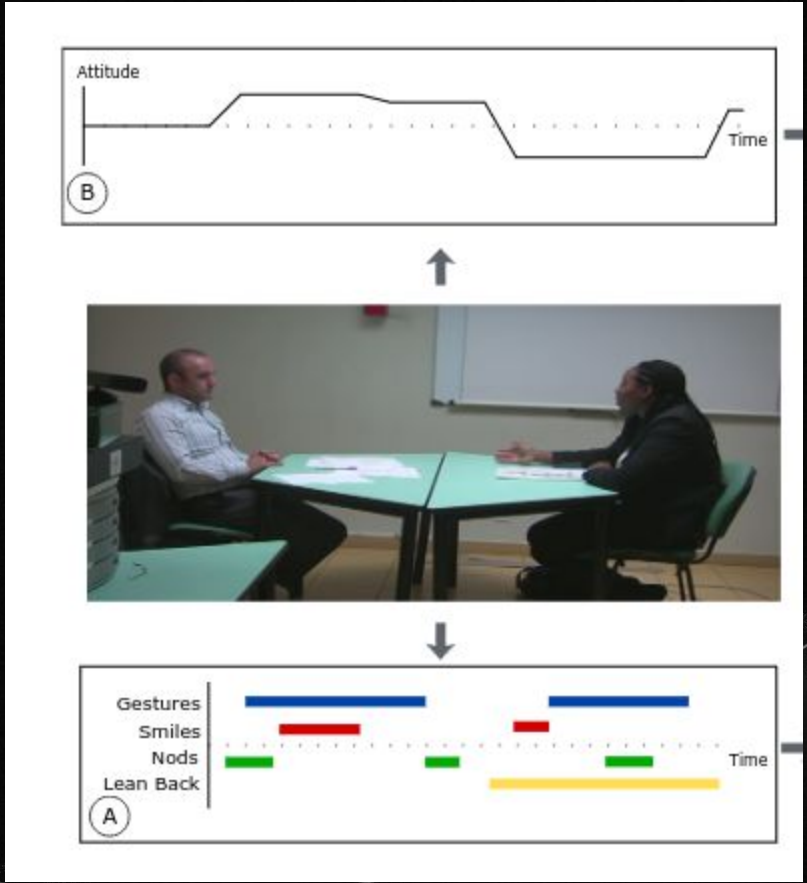


	Gaze			**Eye/Head**	**Blink**	**Nods**
	Direction	*Duration*	*Frequency*	*Direction*	*Frequency*	
Extraversion	Toward	Long	Often	Up, Center, Left, Right	Few	Wide, Fast
Introversion	Avoidance	Short	Seldom	Down, Centre	Normal	Small, Slow
Agreeableness	Toward	Medium	Medium	Centre, Left, Right	Normal	Normal
Non-Agreeableness	Toward	Long	Medium	Down/Up	Few	Head-Shake
Neuroticism	Avoidance	Short	Seldom	Left, Right, Down	Often	None
Emotional Stability	Toward	Medium	Medium	Centre, Left, Right	Normal	None

Ruhland, K., Zibrek, K., & McDonnell, R. (2015). Perception of personality through eye gaze of realistic and cartoon models. In Proceedings of the ACM SIGGRAPH Symposium on Applied Perception (pp. 19-23).



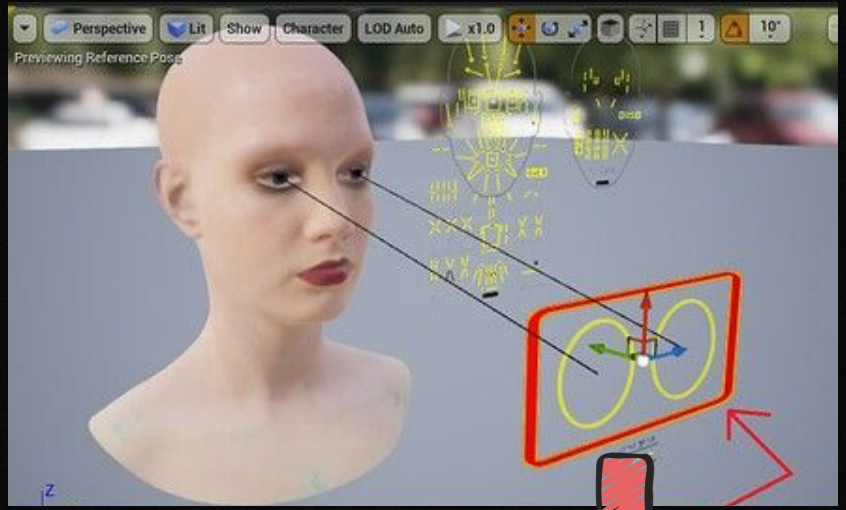
Real-time Multi-map Saliency-driven Gaze Behavior for Non-conversational Characters (Goudé et al., 2023)



A Methodology for the Automatic Extraction and Generation of Non-Verbal Signals Sequences Conveying Interpersonal Attitudes (Chollet et al., 2017)

03

OBJECTIFS DU SEMESTRE 6





Transmettre des émotions, des traits de personnalité ...



… mais aussi : converser (lip sync) et communiquer de façon non verbale

04

ORGANISATION DU GROUPE

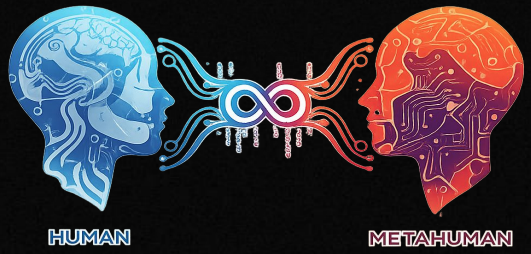
ORGANISATION DU GROUPE

Sous-groupe 1



Introduction de ConvAI dans le projet, réglage du plugin et calibrage.

Sous-groupe 2



Implémentation des émotions et des interactions avec le MetaHuman

05

OUTILS UTILISÉS

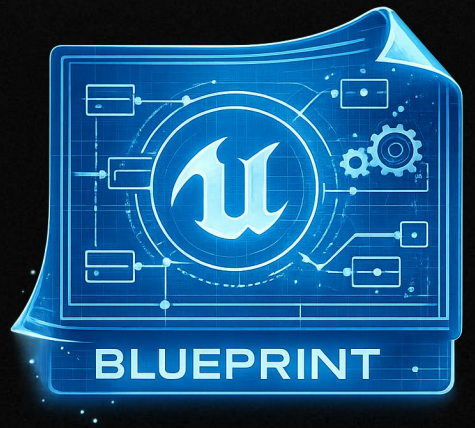
UNREAL ENGINE (V5.7)

- Moteur de jeux
- Création de jeux interactifs et expériences virtuelles
- ***Avantages***:
  robuste, puissant et documentation
- ***Inconvénients***:
  exigences du système, beaucoup à apprendre, fichiers volumineux



BLUEPRINT

- Système de programmation: *script visuel*
- Créé avec C++ par Epic Games
- Pas de code
- Interaction par des nœuds



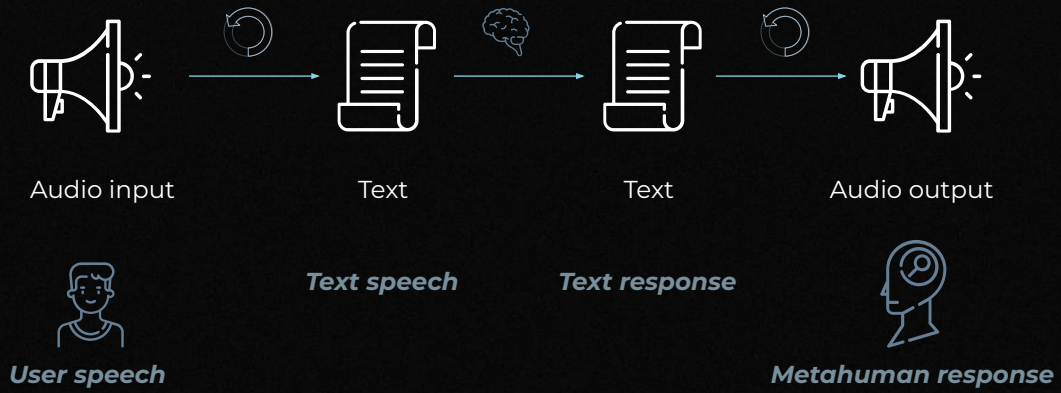
HTC VIVE

- Créé par Valve et HTC
- VIVE + SteamVR = Simulations VR
- Grande immersion
- Indispensable pour eye-tracking





OpenAI API



Interaction MetaHuman-IA

ConvAI

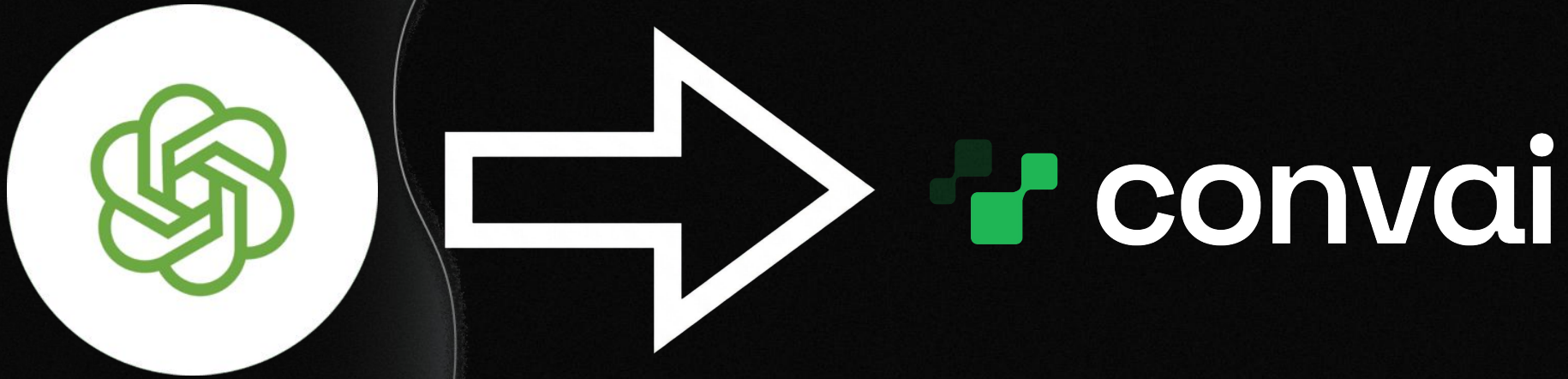


Aussi Intéraction Metahuman - IA

06

PLUGIN ConvAI

De l’API d’OpenAI au plugin ConvAI







Conversation textuelle et vocale

Gratuit



Pas de LIPSYNC

Pas d’animations prédéfinies

Grosse latence entre Questions/Réponses

convai



Petite latence entre Questions/Réponses



Conversation textuelle et vocale

LIPSYNC + Quelques animations de conversations

Ressource en ligne

Gratuit



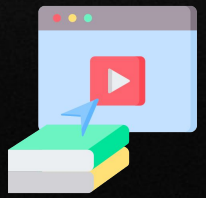
GRATUIT



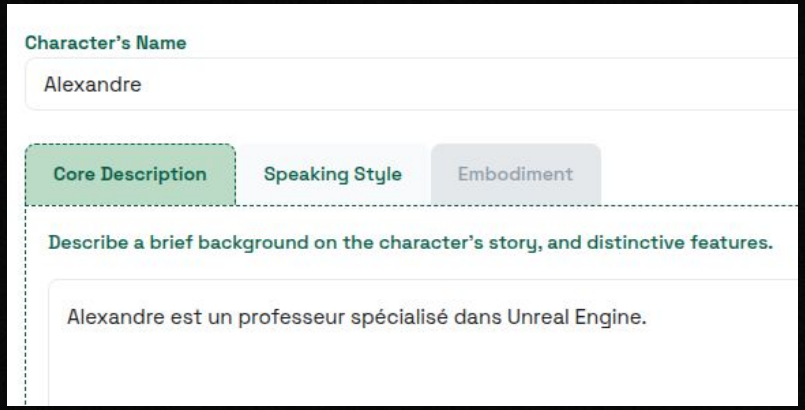
LIPSYNC

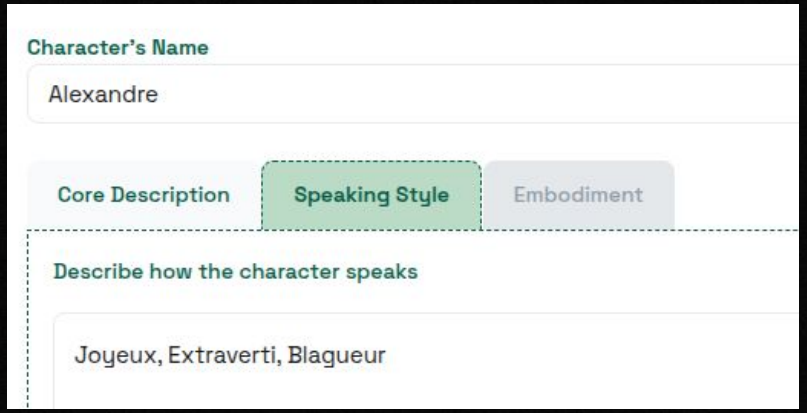


RESSOURCE EN LIGNE



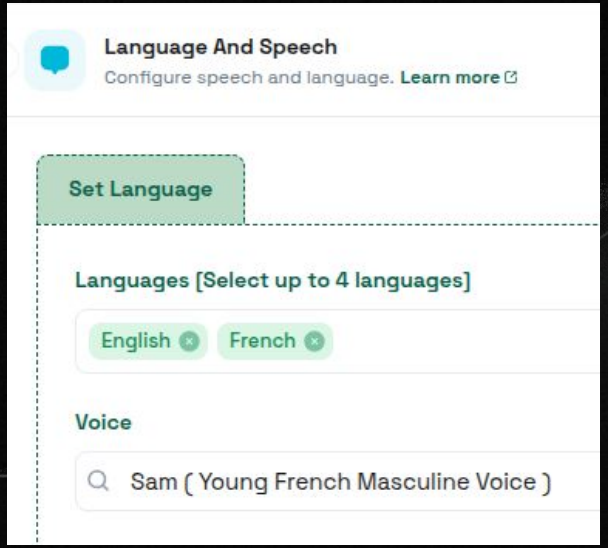
convai



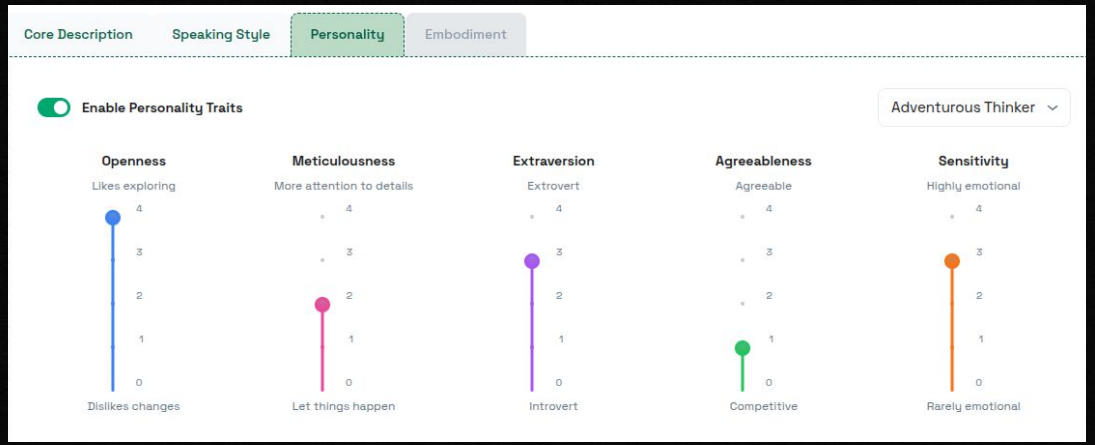


convai

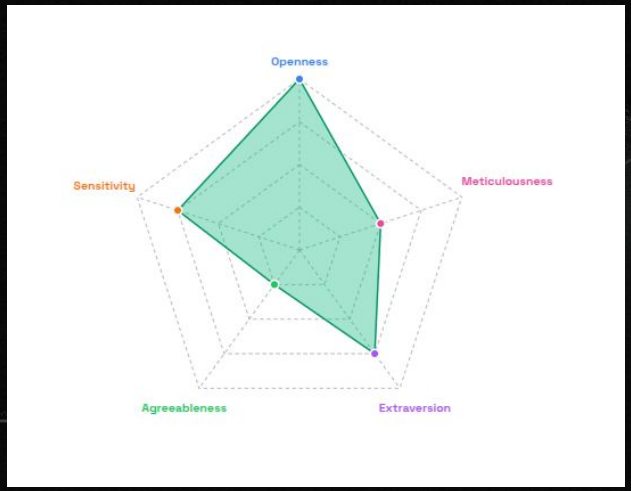
UN PLUGIN TRÈS COMPLET



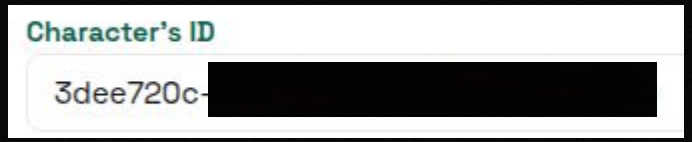


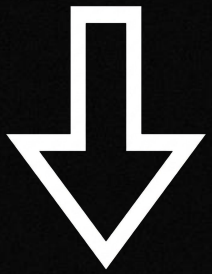


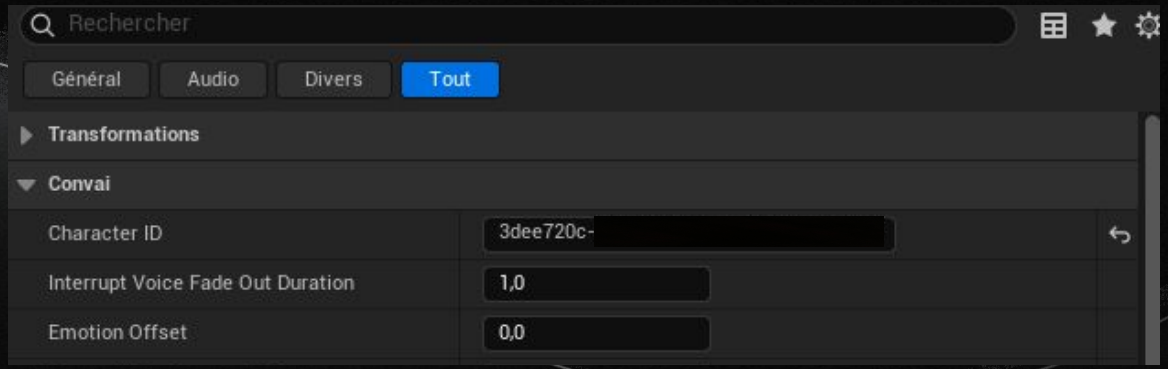
convai



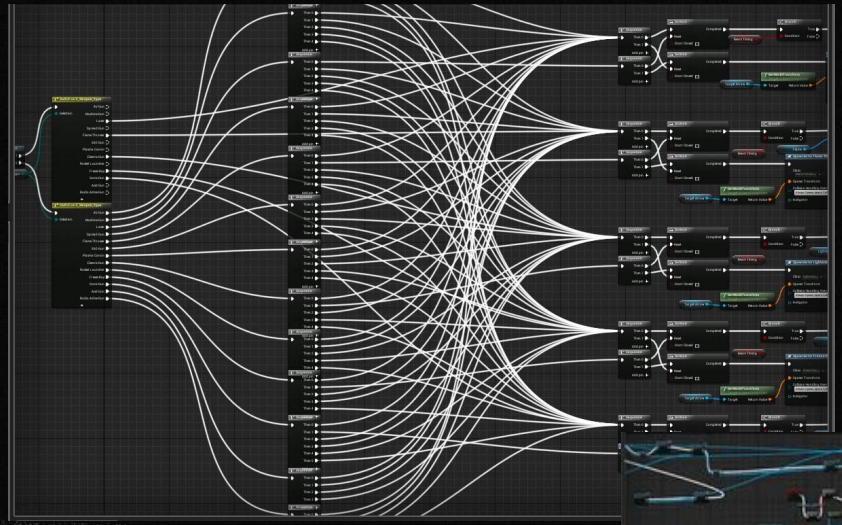
UN PLUGIN TRÈS COMPLET



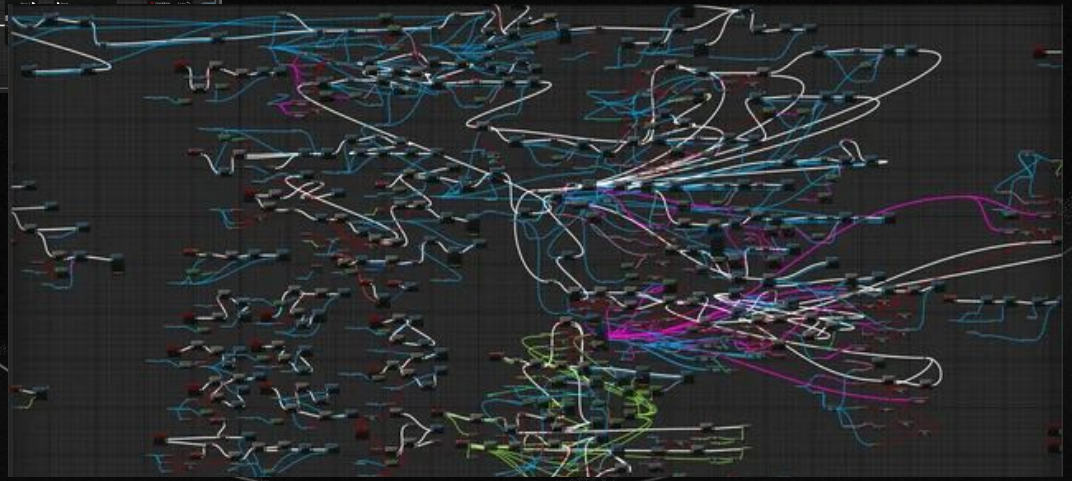




IMPORTATION DU PERSONNAGE DANS UE5







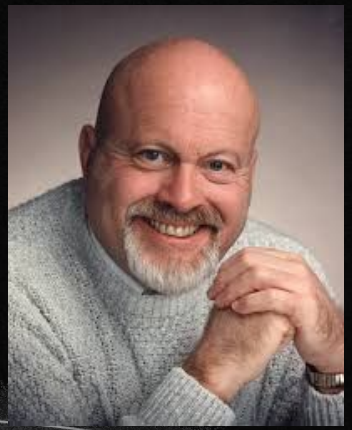


ConvAI, Un plugin IMPOSANT

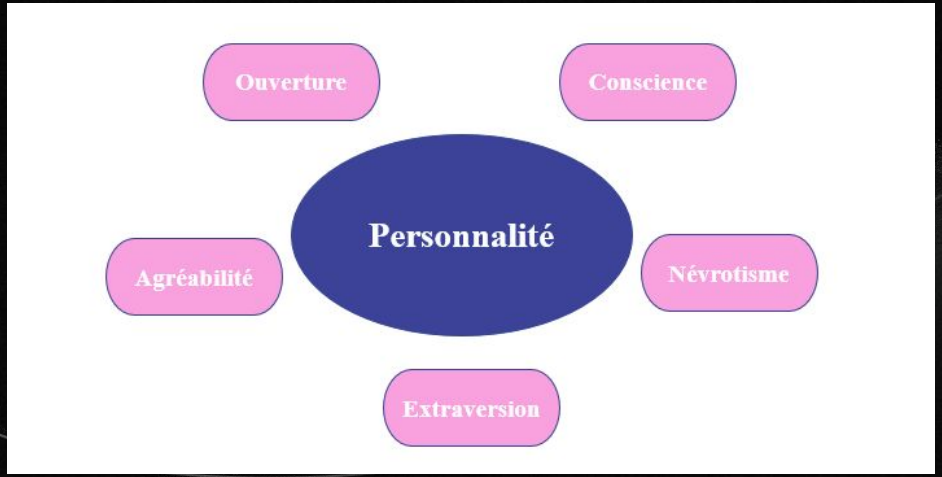
07

DESCRIPTION FONCTIONNELLE

Lewis Goldberg

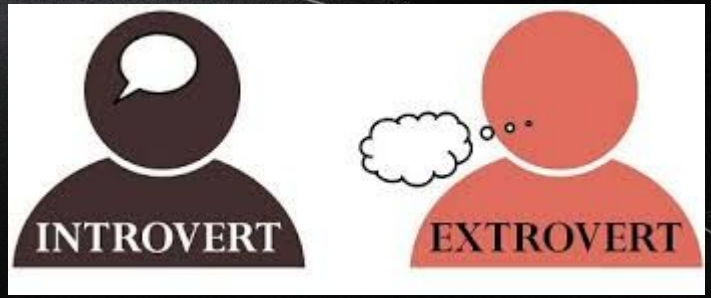


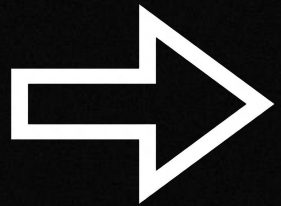




Théorie

	Gaze			**Eye/Head**	**Blink**	**Nods**
	Direction	*Duration*	*Frequency*	*Direction*	*Frequency*	
Extraversion	Toward	Long	Often	Up, Center, Left, Right	Few	Wide, Fast
Introversion	Avoidance	Short	Seldom	Down, Centre	Normal	Small, Slow
Agreeableness	Toward	Medium	Medium	Centre, Left, Right	Normal	Normal
Non-Agreeableness	Toward	Long	Medium	Down/Up	Few	Head-Shake
Neuroticism	Avoidance	Short	Seldom	Left, Right, Down	Often	None
Emotional Stability	Toward	Medium	Medium	Centre, Left, Right	Normal	None



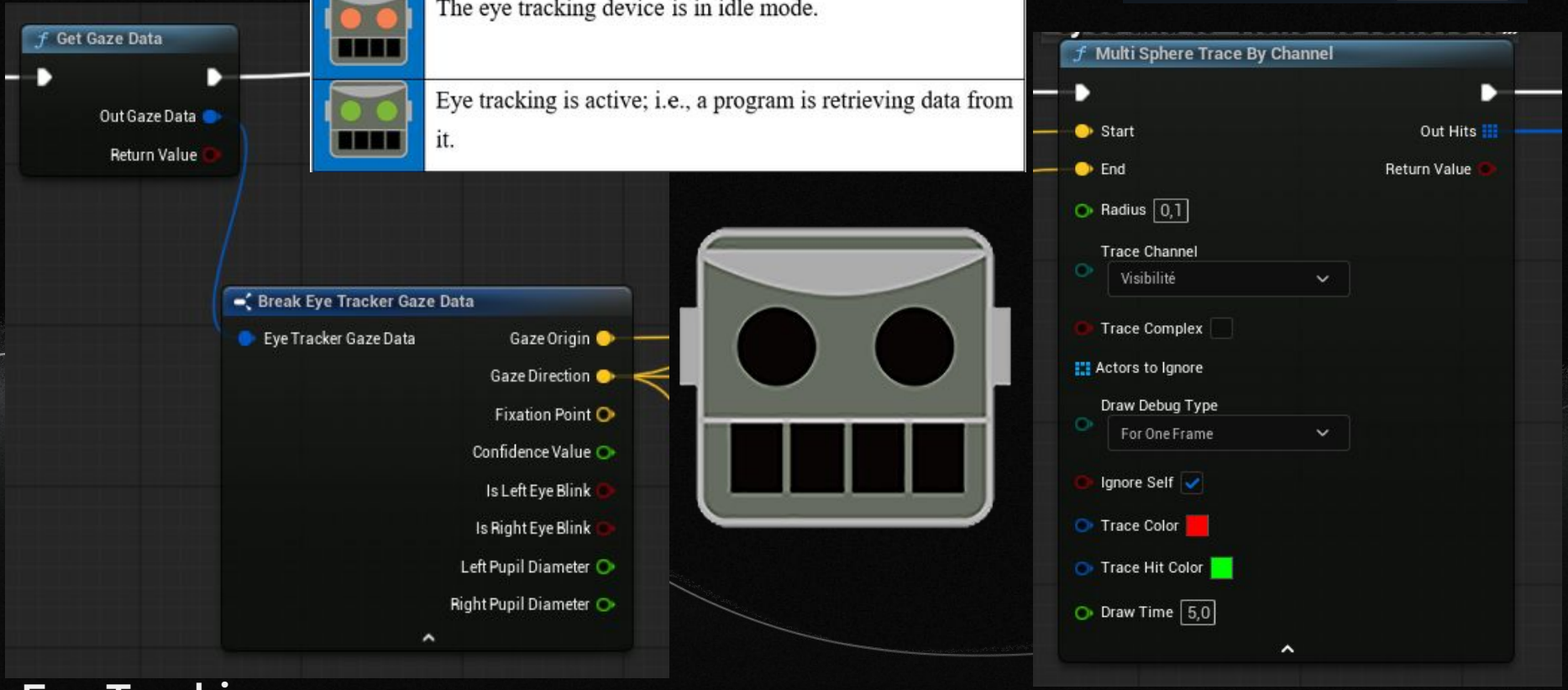




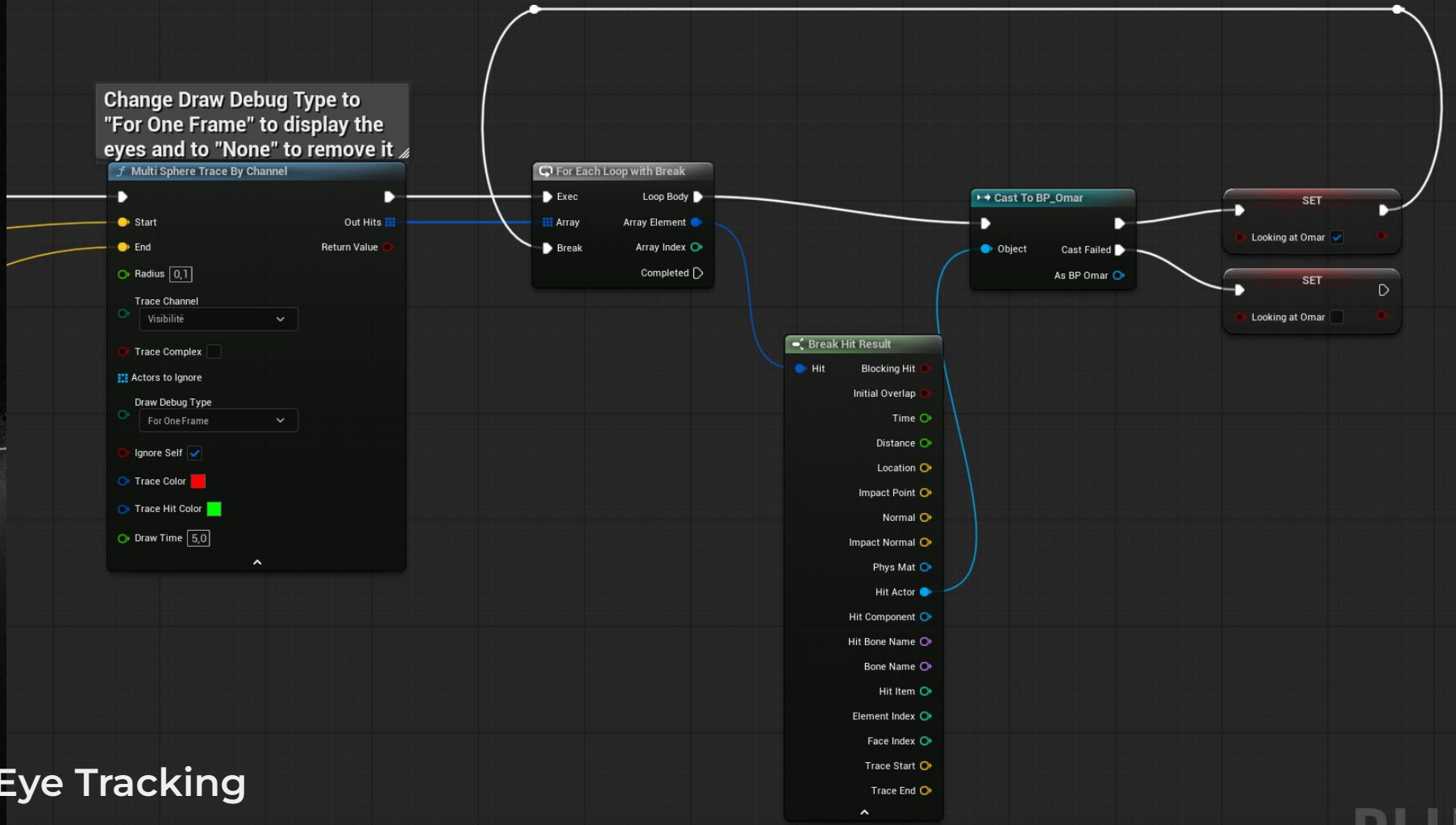
Théorie

	SR runtime is launched but HMD does not support eye tracking or eye tracking has been disabled by the user.
	The eye tracking device is in idle mode.
	Eye tracking is active; i.e., a program is retrieving data from it.

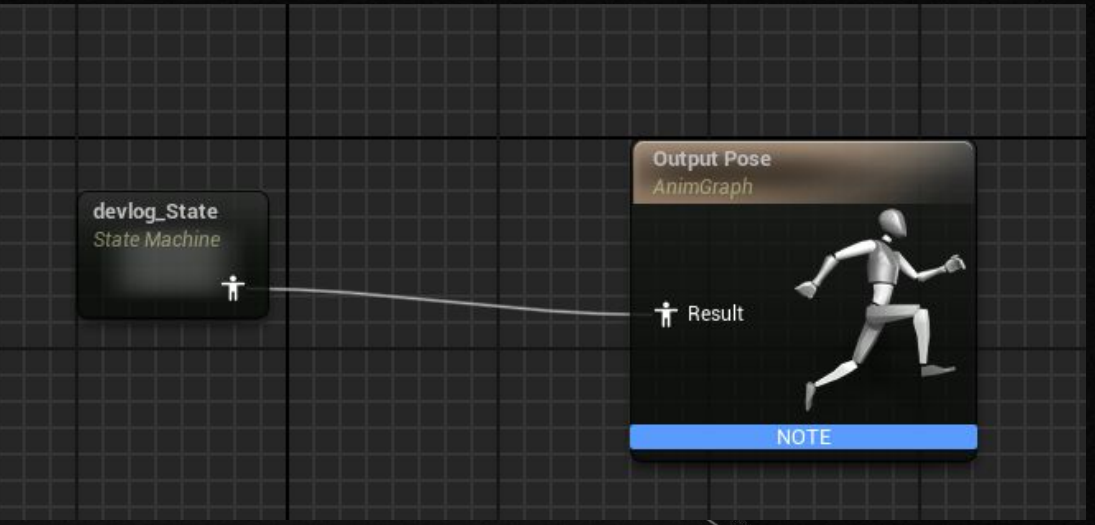
VIVE Console
for SteamVR™

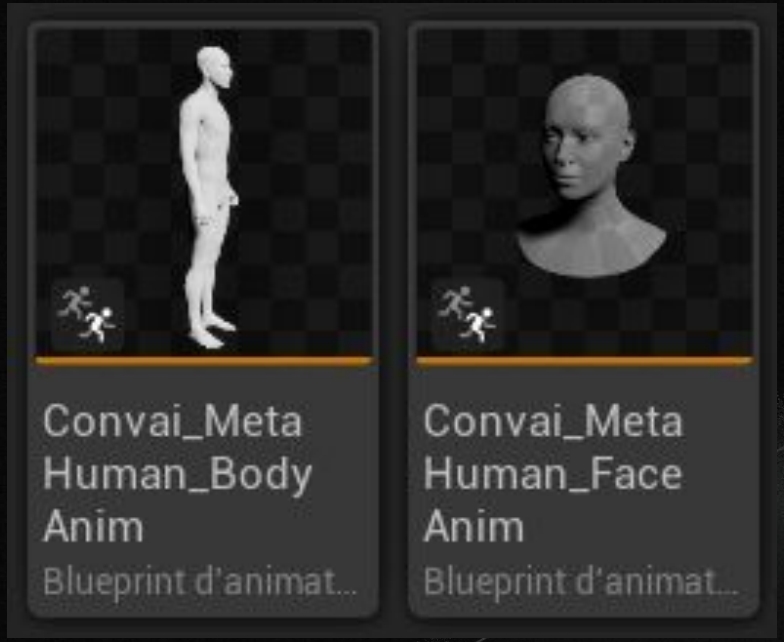


Eye Tracking

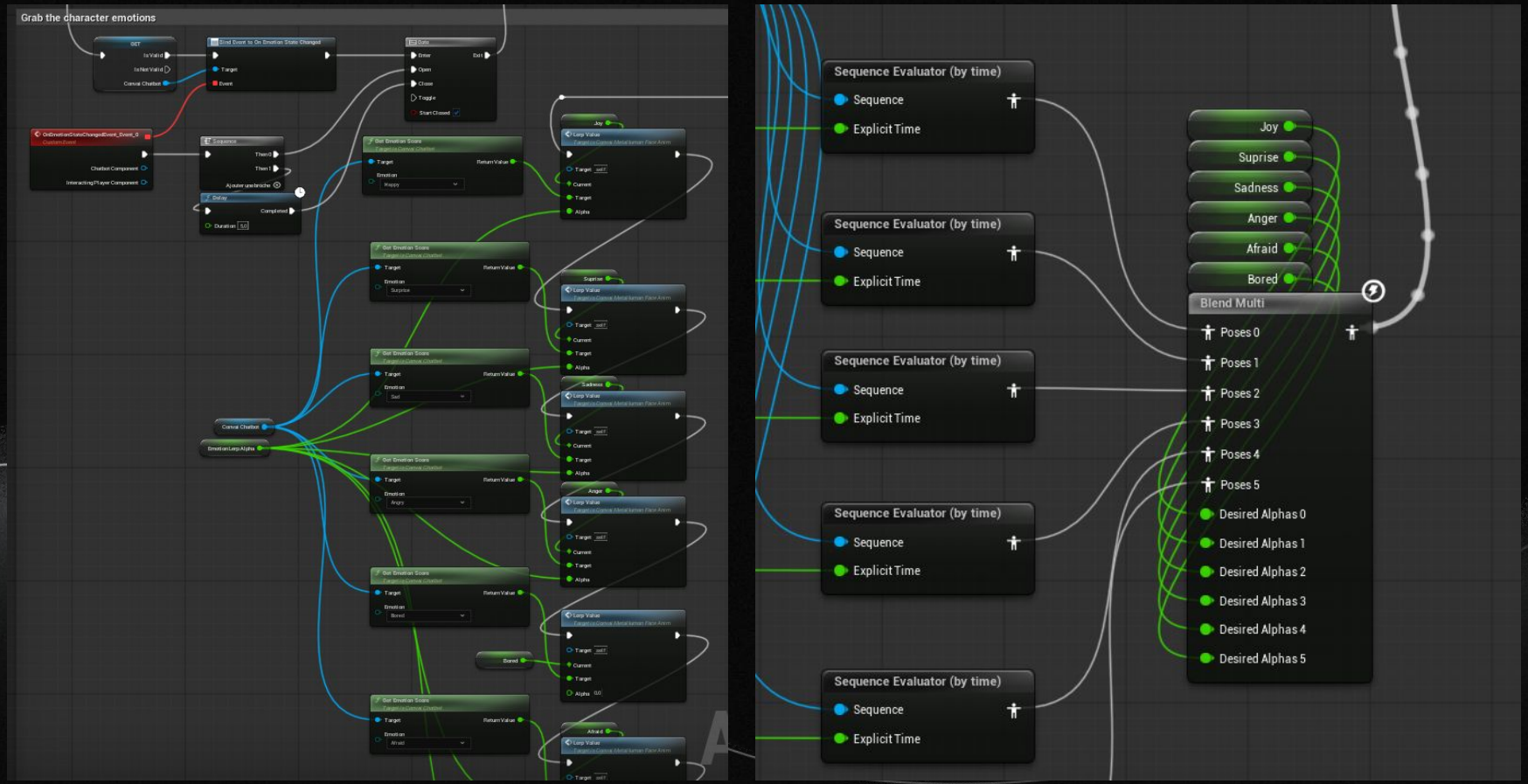


Eye Tracking

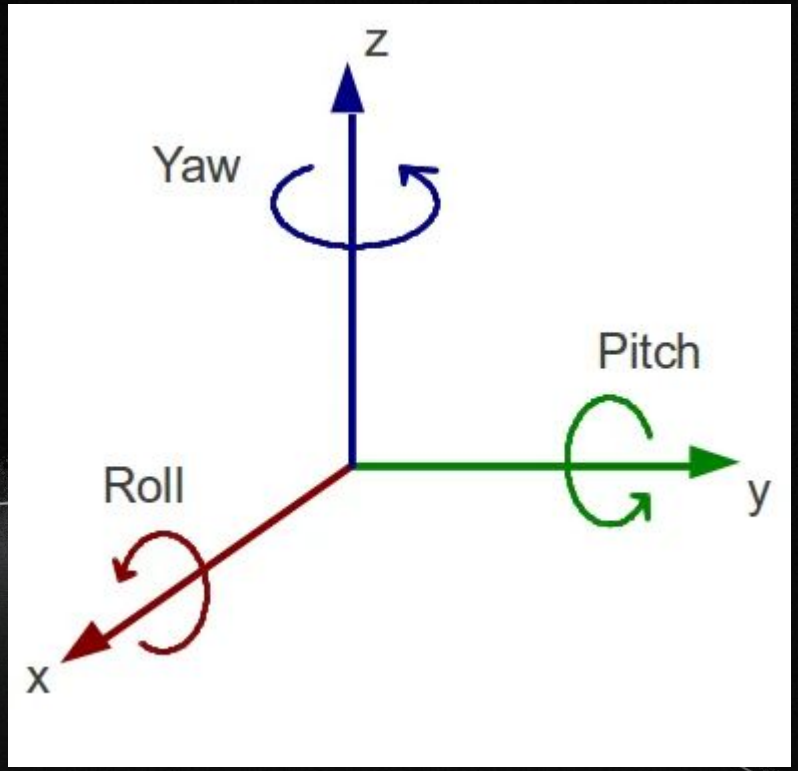


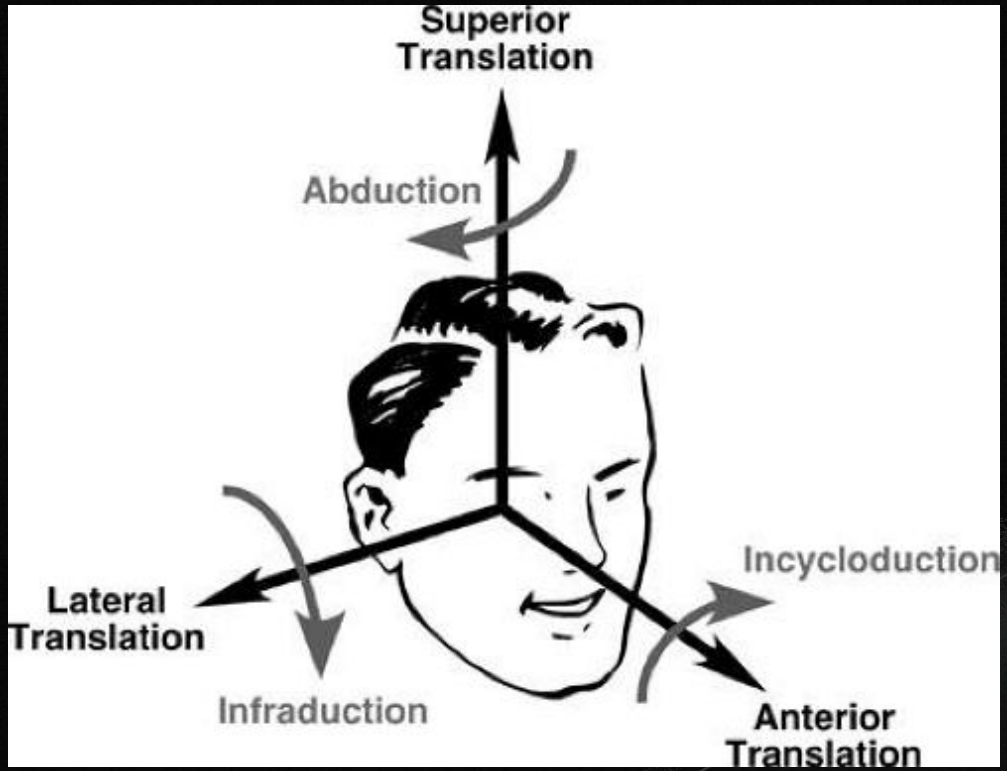


Animation blueprints

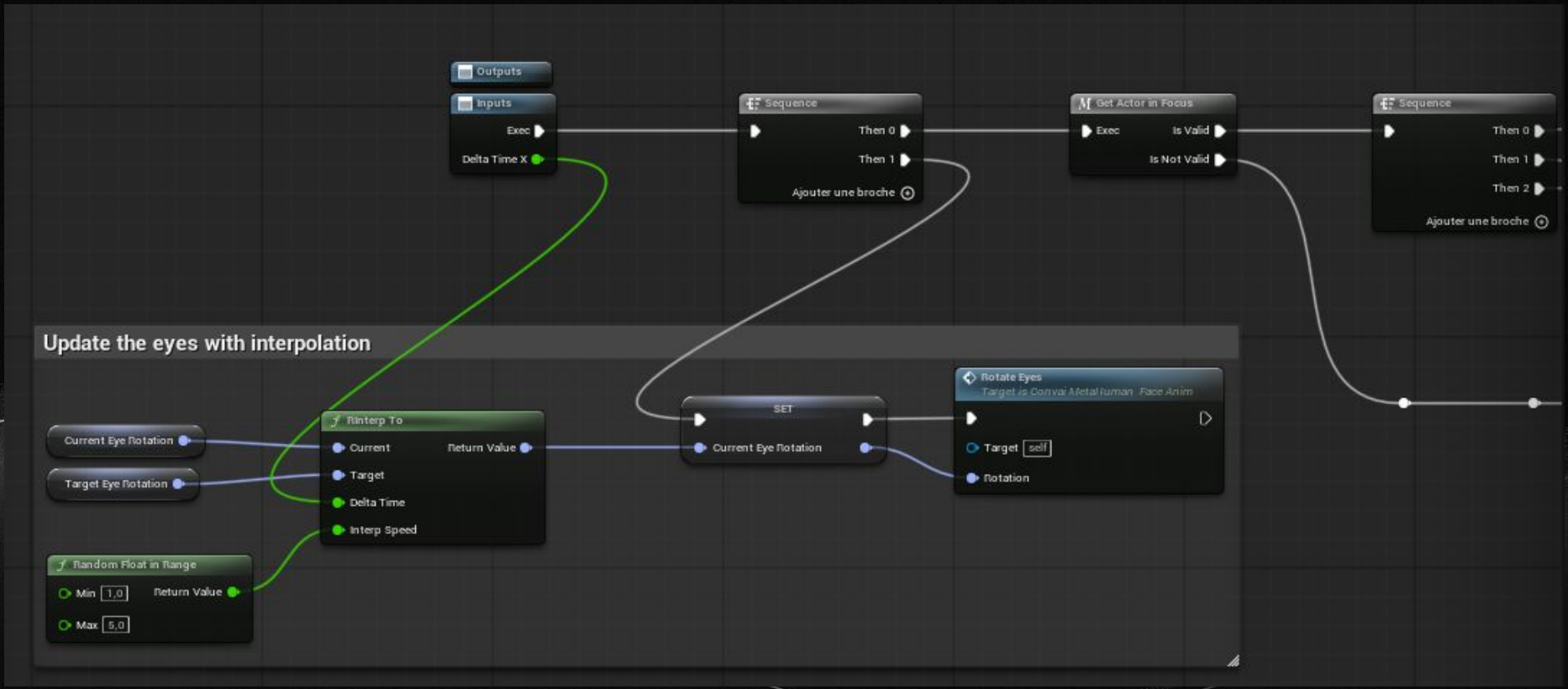

Grab the character emotions
Sequence Evaluator (by time)
Sequence
Explicit Time
Sequence Evaluator (by time)
Sequence
Explicit Time
Sequence Evaluator (by time)
Sequence
Explicit Time
Sequence Evaluator (by time)
Sequence
Explicit Time
Sequence Evaluator (by time)
Sequence
Explicit Time
Joy
Suprise
Sadness
Anger
Afraid
Bored
Blend Multi
Poses 0
Poses 1
Poses 2
Poses 3
Poses 4
Poses 5
Desired Alphas 0
Desired Alphas 1
Desired Alphas 2
Desired Alphas 3
Desired Alphas 4
Desired Alphas 5

Emotions



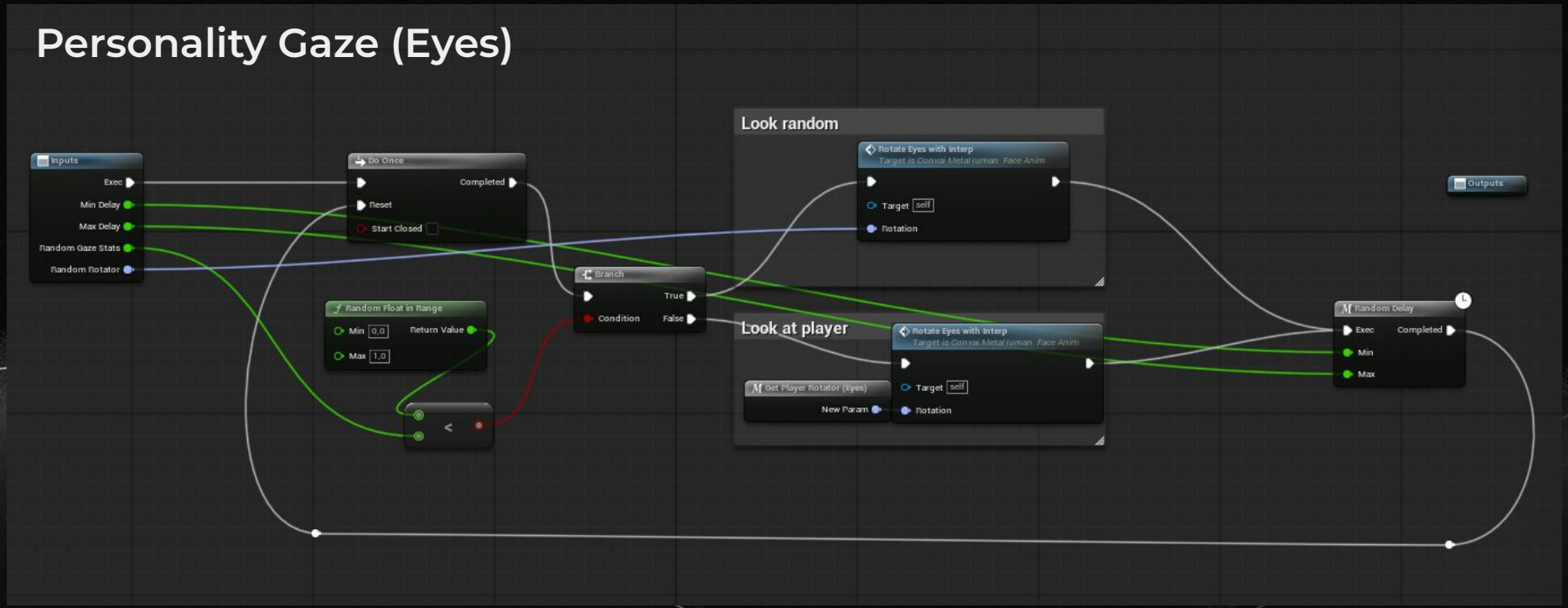


Mouvement des yeux - Rotation

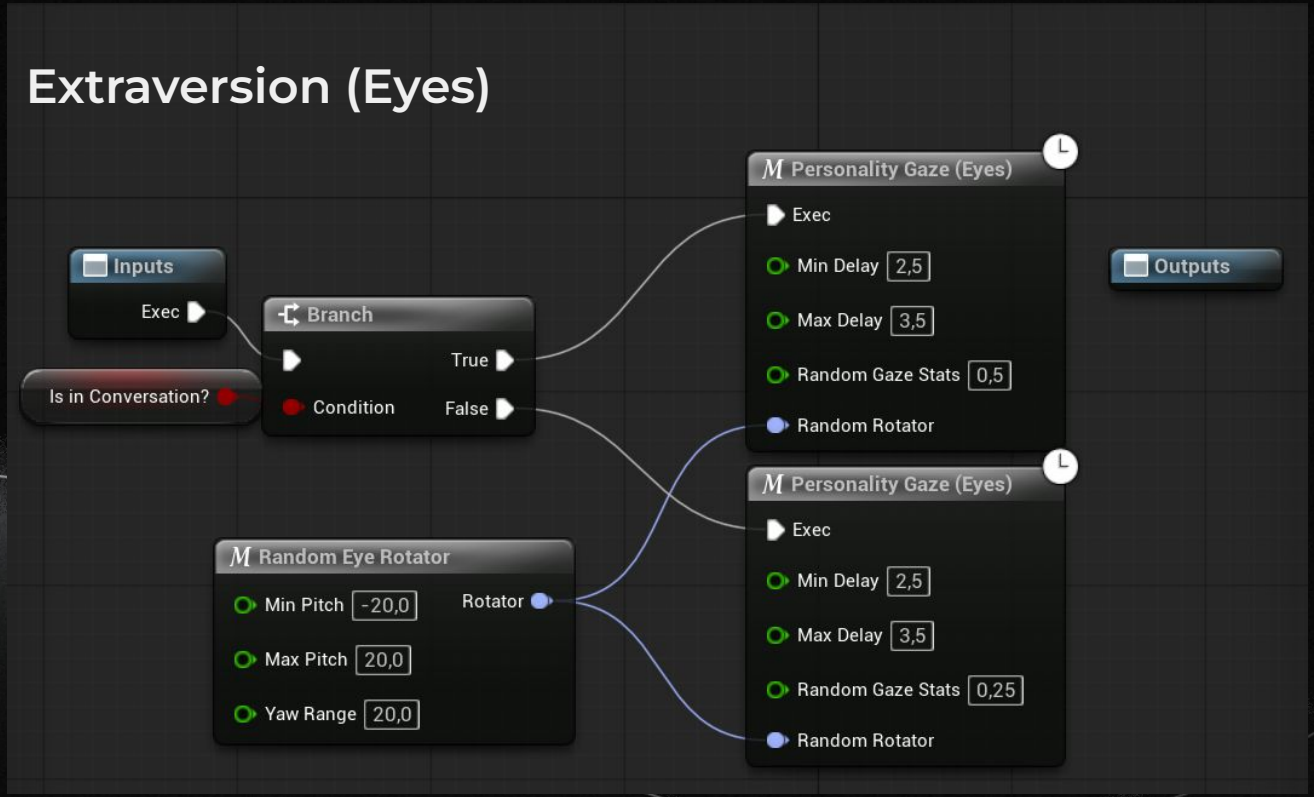


Mouvement des yeux - interpolation

Personality Gaze (Eyes)

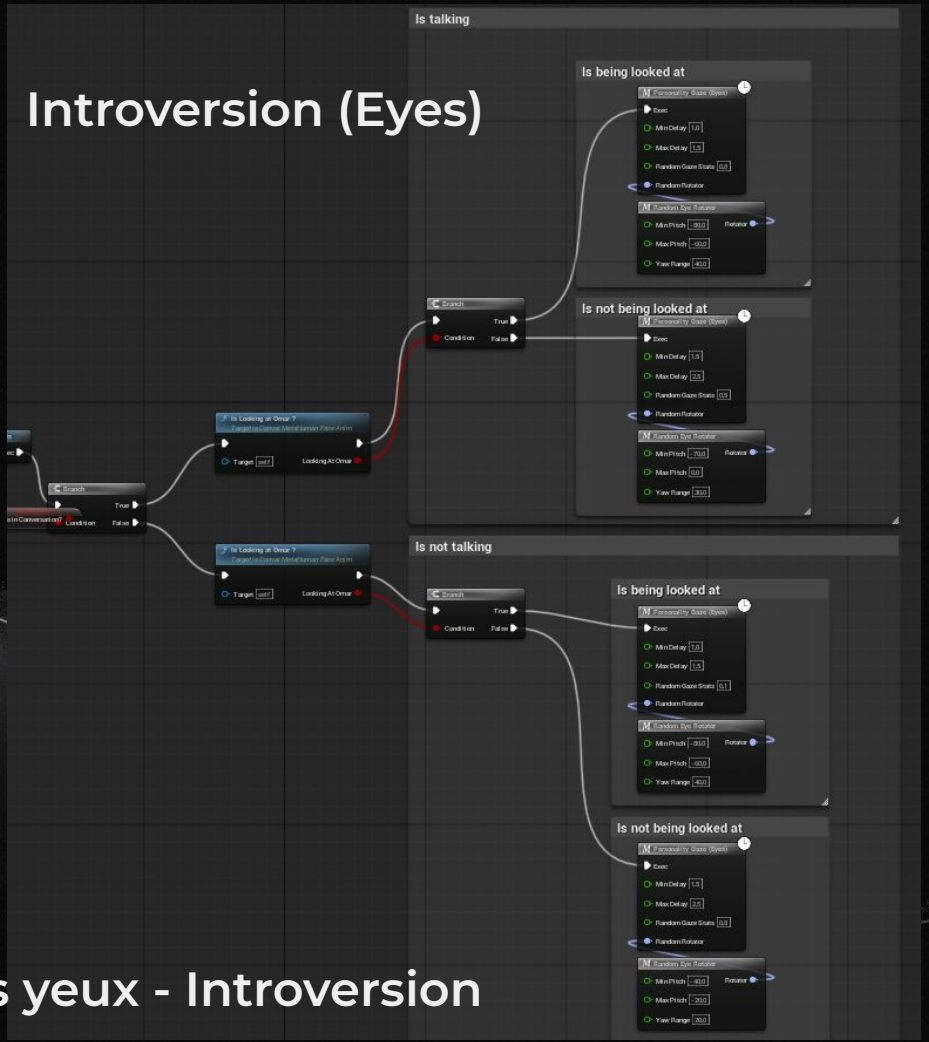


Mouvement des yeux - Personality Gaze

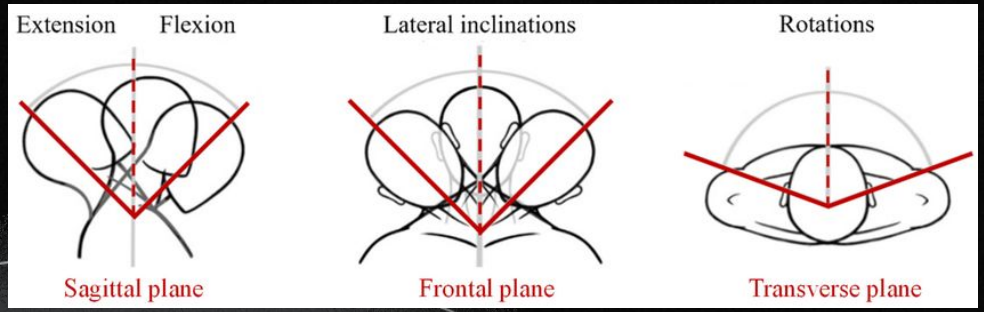


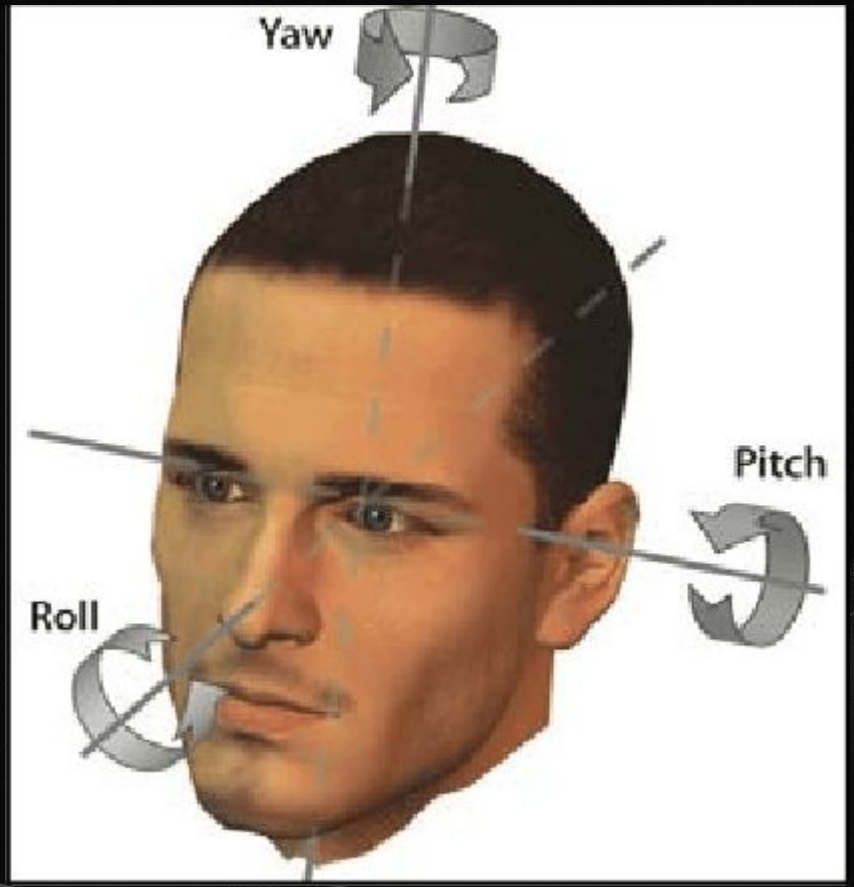
Mouvement des yeux - Extraversion

Introversion (Eyes)



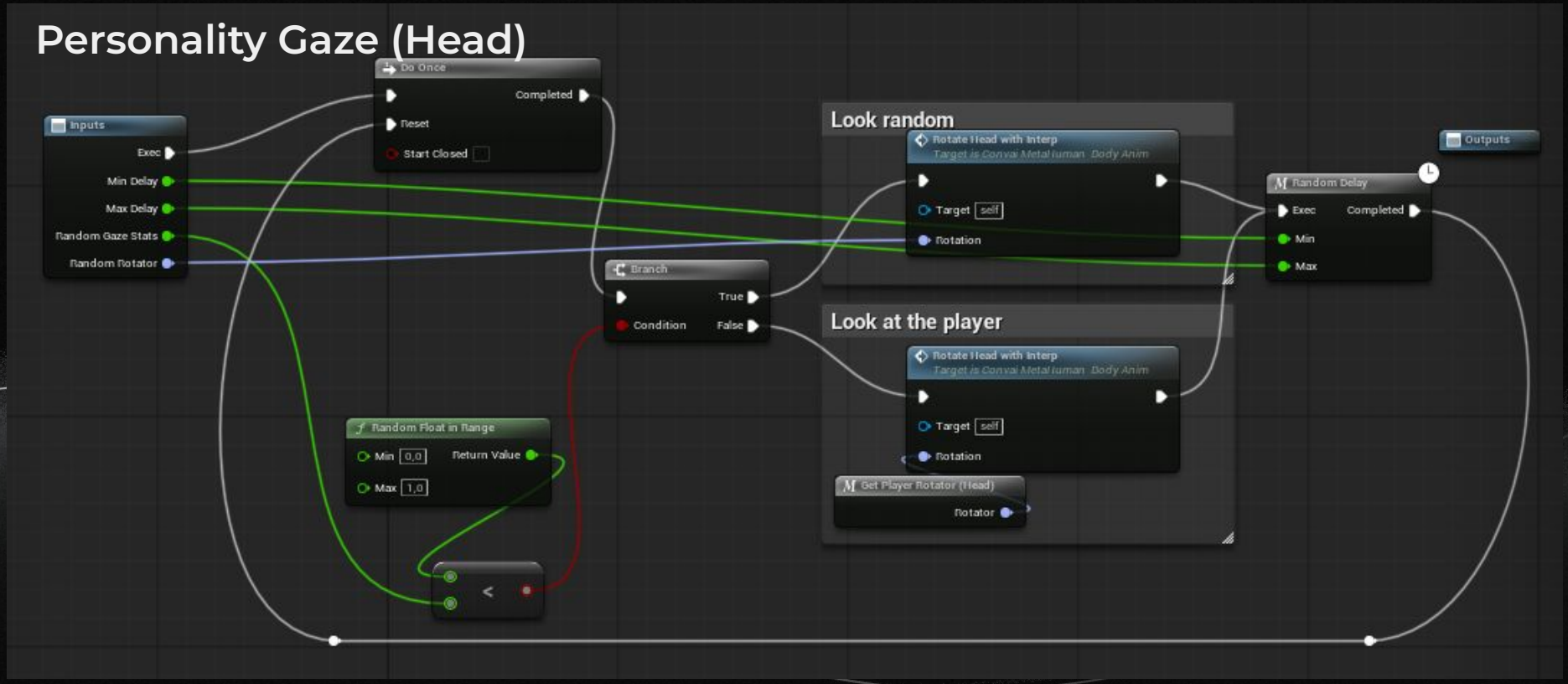
Mouvement des yeux - Introversion



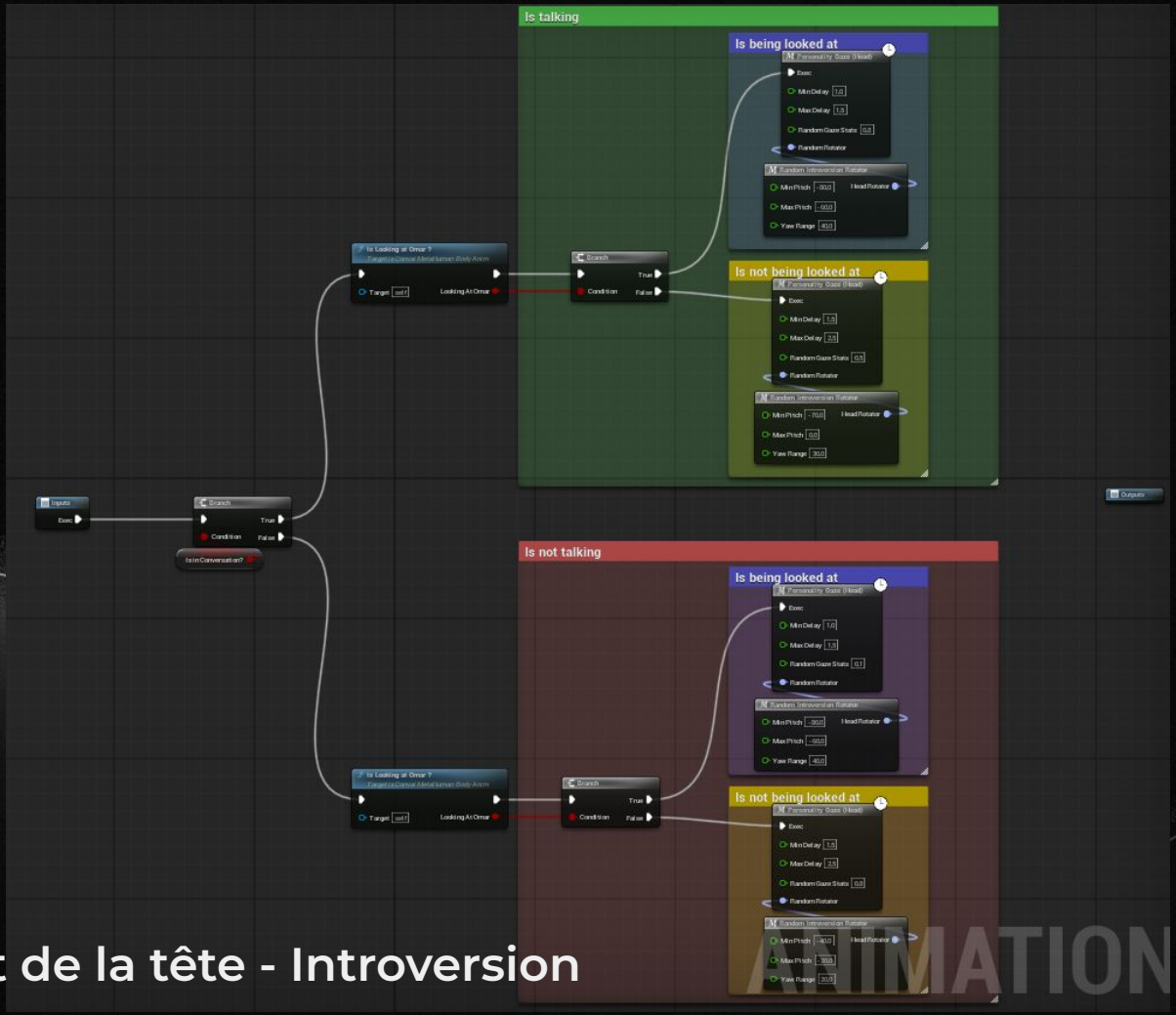


Mouvement de la tête - Personality Gaze

Personality Gaze (Head)



Mouvement de la tête - Personality Gaze

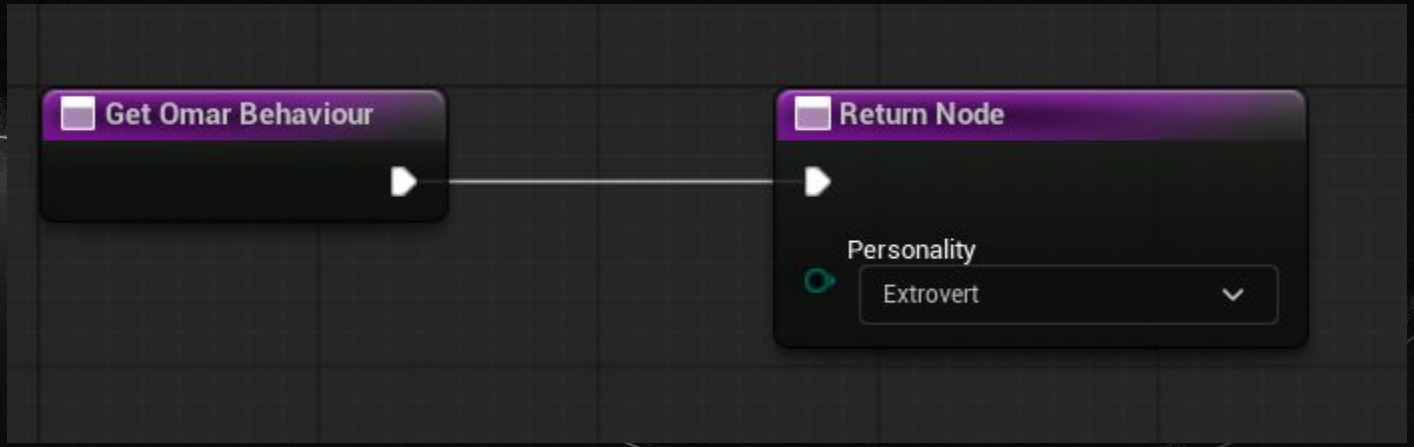


Mouvement de la tête - Introversion

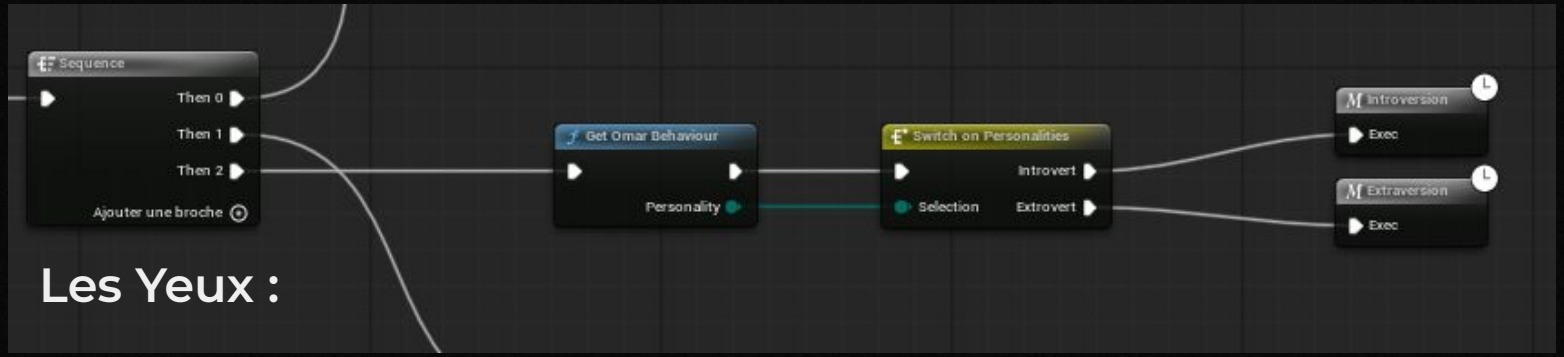


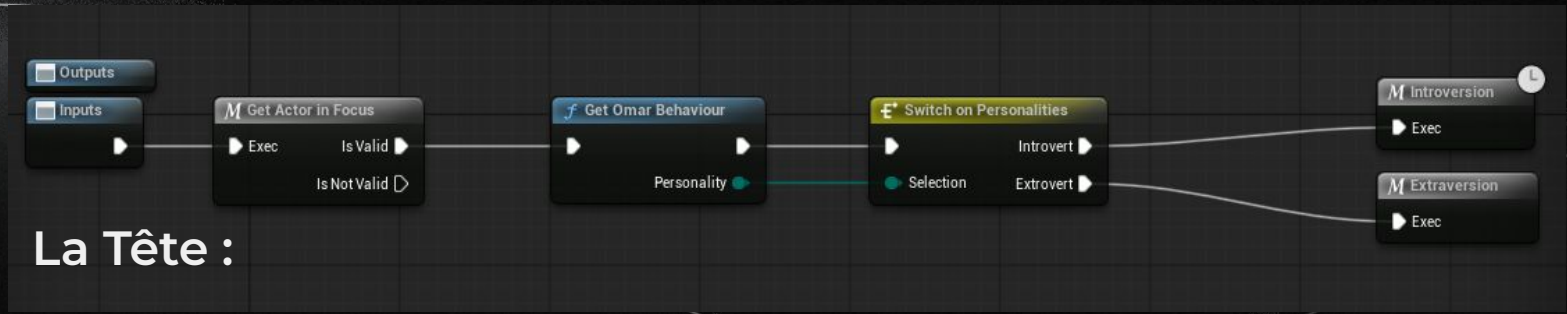
Mouvement de la tête - Introversion



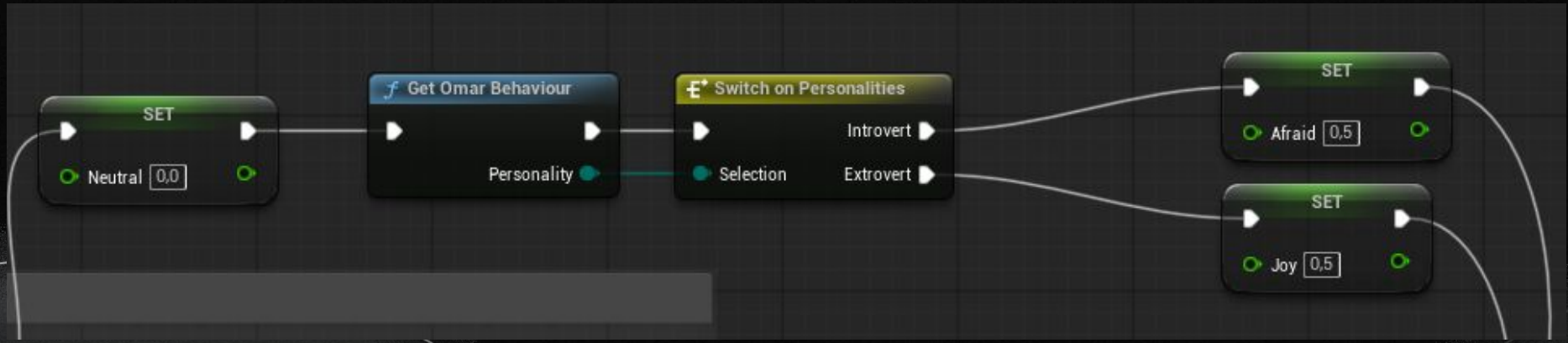


Implémentation : Extraversion / Introversion





Implémentation : Extraversion / Introversion

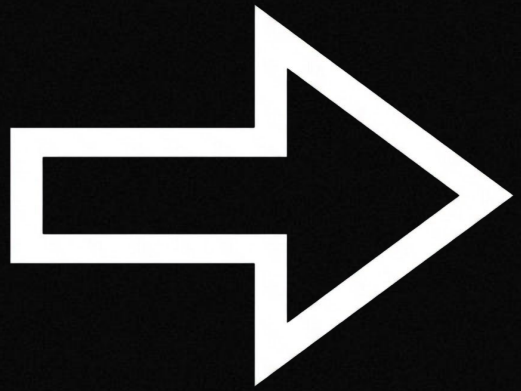


Implémentation : Extraversion / Introversion

08

LES PROBLÈMES RENCONTRÉS





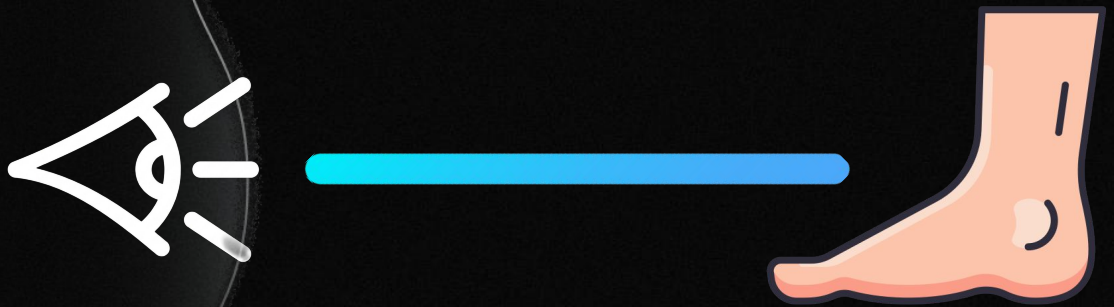
100 requêtes max / mois

convai

Un plugin limité







Problème de calibration HTC Vive





La moitié des animations de ConvAI corrompues

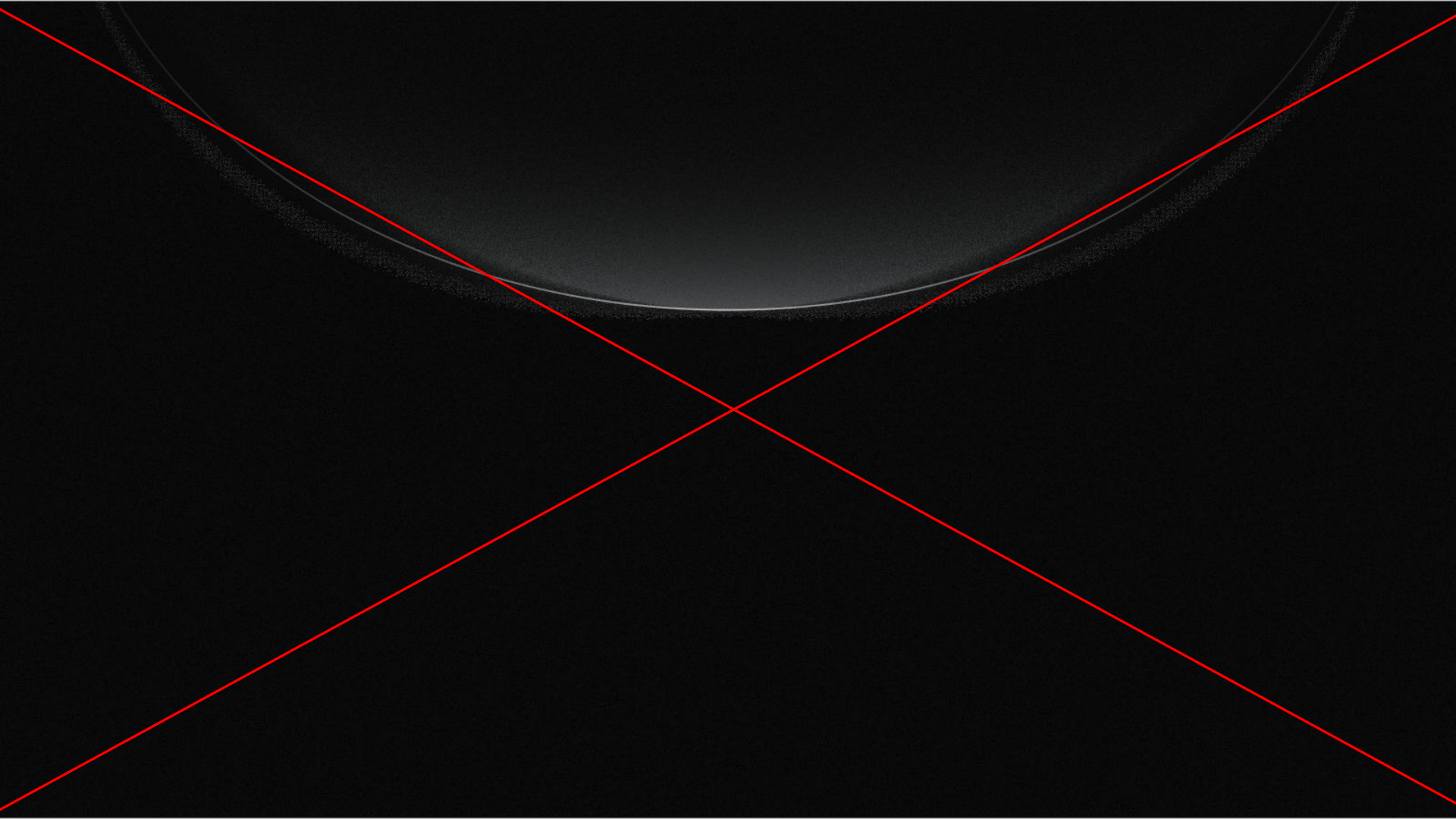
convai :



Modification des fonctionnalités gratuites

09

PROTOTYPE



10

QUELLE EST LA SUITE DE CE PROJET ?

Merci pour votre attention

Bibliographie

[1] S. Sonlu, B. Bendiksen, F. Durupinar et U. Güdükbay : The effects of embodiment and personality expression on learning in llm-based educational agents. arXiv preprint arXiv :2407.10993, 2024.

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